Contents

Constructing the Algebraic Closure **1**

Balasubramanian-Koblitz Theorem **1**
Corollary 1

Reduced Tate **1**

Equivalent Tate Pairing **1**

Frobenius Fixed Points **2**

Tate Trick **2**

Kernel of Map **2**

Restatement of Equivalency **2**

Constructing the Algebraic Closure

Let p be prime and

$$\mathbb{F}_p \subseteq \mathbb{F}_{p^2} \subseteq \cdots \subseteq \mathbb{F}_{p^n} \subseteq \cdots \subseteq \bar{\mathbb{F}}_p$$

$$\bar{\mathbb{F}}_p = \bigcup \mathbb{F}_{p^n}$$

Find a prime poly $f(x) \in \mathbb{F}_p[x]$ (i.e cannot be non-trivially factored such that deg $(f) = n$)

$$\mathbb{F}_p \subseteq \{\mathbb{F}_p[x] \bmod f(x)\} \subseteq \mathbb{F}_p$$

Balasubramanian-Koblitz Theorem

n is prime st. $n \mid \#E(\mathbb{F}_p)$ and $\gcd(n, p-1) = 1$. Then

$$E[n] \subseteq E(\mathbb{F}_{p^k}) \iff n \mid p^k - 1$$

The embedding degree of (E, n) is the minimal k st. $n \mid p^k - 1$.

Corollary

k is embedding degree of E, then $\mu_n \subseteq \mathbb{F}_{p^k}$.

Reduced Tate

$$\tau_n(P, Q) = f_{nD_P}(D_Q)^{\frac{p^k-1}{n}}$$

Equivalent Tate Pairing

Let Q_0 be such that $nQ_0 = Q$. Such a point is guaranteed to exist by the surjectivity of multiplication by n map.

$$\begin{array}{ccc} Q_0 & \longrightarrow & E(\mathbb{F}_{p^k}) \ni Q \\ \downarrow & & \downarrow \\ n : E(\overline{\mathbb{F}_{p^k}}) & \longrightarrow & E(\overline{\mathbb{F}_{p^k}}) \end{array}$$

Then let $Q_1 = (\Phi^k - 1)(Q_0)$ where $Q_0 \in E[n]$ and Φ is the frobenius automorphism. Then $Q_1 \in E[n]$.

$$\begin{aligned} e_n(P, Q_1) &= \frac{g_P(S + Q_1)}{g_P(S)} \\ &= \tau_n(P, Q) \end{aligned}$$

Frobenius Fixed Points

$$\Phi : \overline{\mathbb{F}_p} \to \overline{\mathbb{F}_p}$$

Abuse of notation:

$$\begin{gathered} \Phi : E(\overline{\mathbb{F}_p}) \to E(\overline{\mathbb{F}_p}) \\ \text{FixedPoints}(\Phi) = E(\mathbb{F}_p) \\ \Phi^k : \overline{\mathbb{F}_{p^k}} \to \overline{\mathbb{F}_{p^k}} \\ \Phi^k : E(\overline{\mathbb{F}_{p^k}}) \to E(\overline{\mathbb{F}_{p^k}}) \\ \text{FixedPoints}(\Phi^k) = E(\mathbb{F}_{p^k}) \end{gathered}$$

Tate Trick

$$\frac{\phi^k - 1}{n} : E(\mathbb{F}_{p^k}) \to E[n]$$

but $Q \in E(\mathbb{F}_{p^k})$

$$\begin{gathered} Q = nQ_0 \to (\Phi^k - 1)(Q_0) = Q_1 \\ Q_0 \in E(\overline{\mathbb{F}_{p^k}}) \end{gathered}$$

Kernel of Map

$$\ker\left(\frac{\Phi^k - 1}{n}\right) = nE(\mathbb{F}_{p^k}) \subseteq E(\mathbb{F}_{p^k})$$

if $Q = nP \in E(\mathbb{F}_{p^k})$

$$\begin{aligned} \left(\frac{\Phi^k - 1}{n}\right)(nP) &= (\Phi^k - 1)(P) \\ &= \Phi^k P - P \end{aligned}$$

Restatement of Equivalency

$$\begin{gathered} \forall P \in E[n] \subseteq E(\mathbb{F}_{p^k}) \\ \forall Q \in E(\mathbb{F}_{p^k}) \\ Q_1 = \left(\frac{\Phi^k - 1}{n}\right)(Q) \\ \implies \tau_n(P, Q) = e_n(P, Q_1) \end{gathered}$$

By 1st isomorphism theorem

$$\begin{gathered} \frac{\Phi^k - 1}{n} : E(\mathbb{F}_{p^k}) \to E[n] \\ E(\mathbb{F}_{p^k})/nE(\mathbb{F}_{p^k}) \cong \text{Im}\left(\frac{\Phi^k - 1}{n}\right) \end{gathered}$$